

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Aashif Rahman
Program/Batch	B083-Networkwalks
Date	25 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	Networkwalks domain (lab-assigned scope) Local lab network / authorized scan targets
Permission secured from client?	As per the assigned Networkwalks internship lab scope
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I performed these activities only within the systems and devices included in the assigned educational lab scope or on systems for which I had appropriate authorization. The material in this report is intended for cybersecurity education and research. Reconnaissance and scanning should not be performed against third-party systems without permission. The findings below are observations from the exercises and are not, by themselves, proof of exploitable vulnerabilities.

2. Introduction

This report documents my Week 2 practical activities covering footprinting/reconnaissance of the networkwalks.com domain and network discovery using Zenmap. The work demonstrates how security professionals collect publicly available information, fingerprint web technologies, inspect DNS and HTTP responses, identify defensive controls, and perform controlled network discovery.

The evidence in this report is based on the screenshots from my Kali Linux and Zenmap lab sessions. Each activity includes the command used, the observed result, and the security relevance of the observation.

3. Tools Used

Tool	Purpose
Kali Linux	Operating system used for reconnaissance and security tooling.
WHOIS	Collect public domain registration and name-server information.
WhatWeb	Fingerprint web technologies, server details and application components.
nslookup	Resolve the domain name through DNS.
curl -I	Inspect HTTP response headers.
wafw00f	Identify the presence/type of a Web Application Firewall.
DNSRecon	Enumerate DNS records such as NS, MX, A, TXT and SRV.
Zenmap (Nmap GUI)	Perform host/service discovery and visualize network topology.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I used six Kali Linux tools to collect different categories of information from networkwalks.com: WHOIS, WhatWeb, nslookup, curl, wafw00f and DNSRecon. The results were used for information gathering and fingerprinting rather than exploitation.

1. WHOIS
Command: whois networkwalks.com
Observation: The output returned registration metadata, registrar information, name servers, domain status and DNSSEC status.

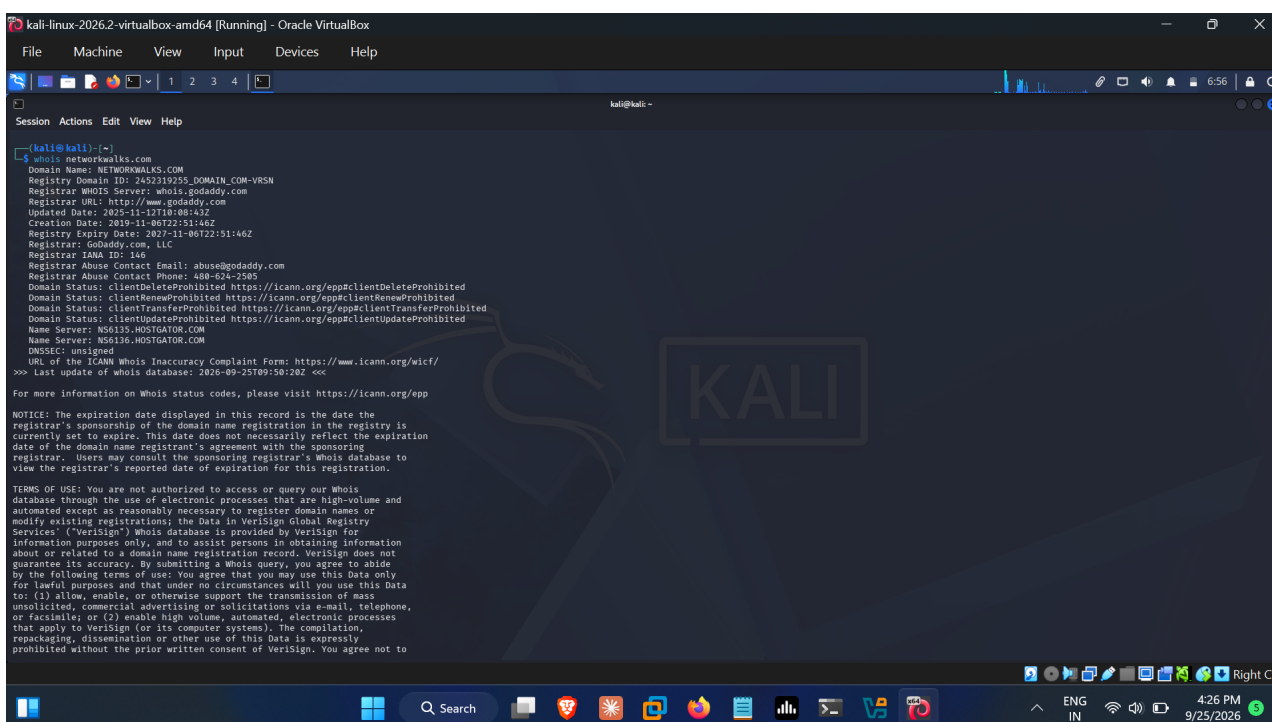
2. WhatWeb
Command: whatweb networkwalks.com
Observation: The screenshot identified Apache, the domain IP 192.232.216.135, WordPress 7.1.2, WordPress Download Manager 3.3.58, jQuery 3.7.1 and other web technologies/headers.

3. Nslookup
Command: nslookup networkwalks.com
Observation: DNS resolution returned networkwalks.com at 192.232.216.135 using the DNS server 8.8.8.8.

4. Curl
Command: curl -I https://networkwalks.com
Observation: The response was HTTP/2 200 and exposed standard response-header information, including a WordPress REST API link under /wp-json/.

5. Wafw00f
Command: wafw00f networkwalks.com
Observation: The tool reported that the site is behind ModSecurity (SpiderLabs) WAF and completed two requests.

6. DNSRecon
Command: dnsrecon -d networkwalks.com
Observation: The output enumerated DNS information including SOA/NS, MX, A, TXT and SRV records. The screenshot also shows a DNSSEC query with no answer.



```
kali@kali:~$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp/clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp/clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp/clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp/clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-25T09:50:20Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
```

Evidence 1 — WHOIS output for networkwalks.com.

```
aashif@AAS43:~$ sudo apt-get install ruby-rchardet
Get:4 http://kali.download/kali kali-last-snapshot/main amd64 ruby-rchardet all 1.10.2-1 [92.1 kB]
Fetched 312 kB in 1s (227 kB/s)
Selecting previously unselected package ruby-public-suffix.
(Reading database... 136399 files and directories currently installed.)
Preparing to unpack ./ruby-public-suffix_7.0.5-1_all.deb...
Unpacking ruby-public-suffix (7.0.5-1)...
Selecting previously unselected package ruby-addressable.
Preparing to unpack ./ruby-addressable_2.9.0-1_all.deb...
Unpacking ruby-addressable (2.9.0-1)...
Selecting previously unselected package ruby-ipaddress.
Preparing to unpack ./ruby-ipaddress_0.8.3-6_all.deb...
Unpacking ruby-ipaddress (0.8.3-6)...
Selecting previously unselected package ruby-rchardet.
Preparing to unpack ./ruby-rchardet_1.10.2-1_all.deb...
Unpacking ruby-rchardet (1.10.2-1)...
Selecting previously unselected package whatweb.
Preparing to unpack ./whatweb_0.6.4-1_all.deb...
Unpacking whatweb (0.6.4-1)...
Setting up ruby-public-suffix (7.0.5-1)...
Setting up ruby-ipaddress (0.8.3-6)...
Setting up ruby-rchardet (1.10.2-1)...
Setting up ruby-addressable (2.9.0-1)...
Setting up whatweb (0.6.4-1)...
Processing triggers for man-db (2.13.1-1)...

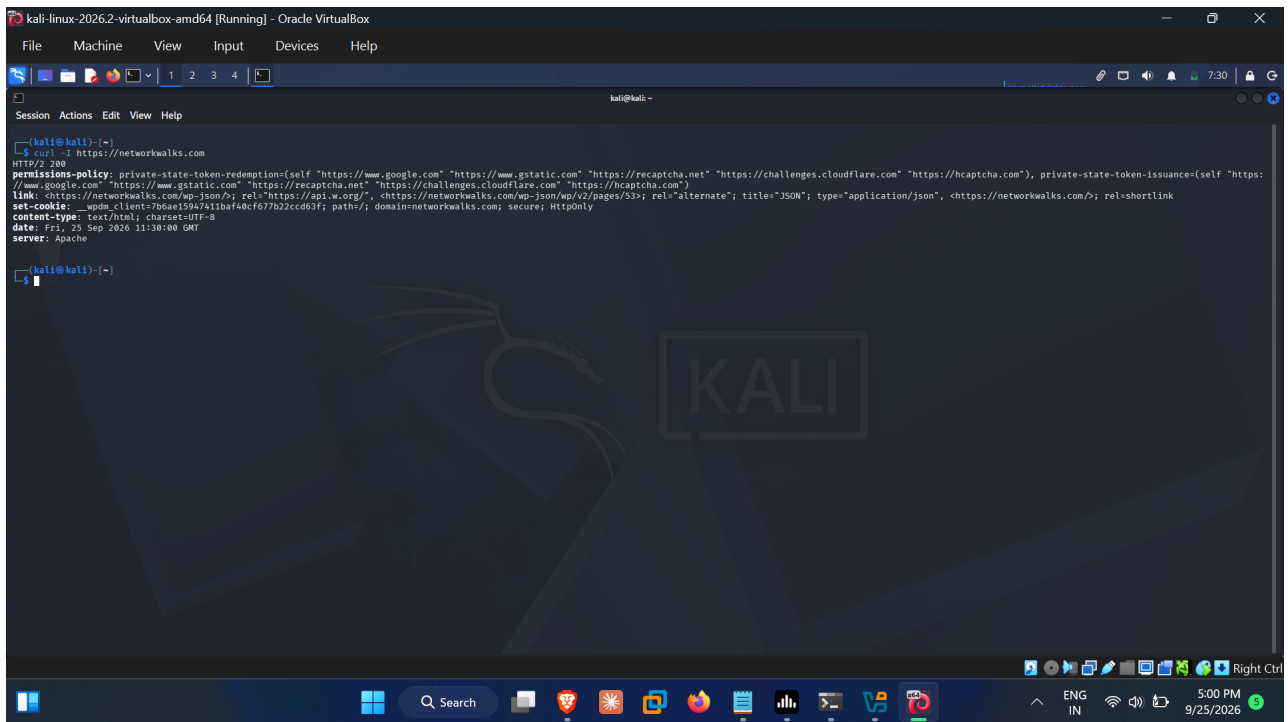
aashif@AAS43:~$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[_wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.2], Cookies[_wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.2,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade], WordPress[7.1.2]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.2], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.2,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade], WordPress[7.1.2]
```

Evidence 2 — WhatWeb fingerprinting output for networkwalks.com.

```
kali@kali:~$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

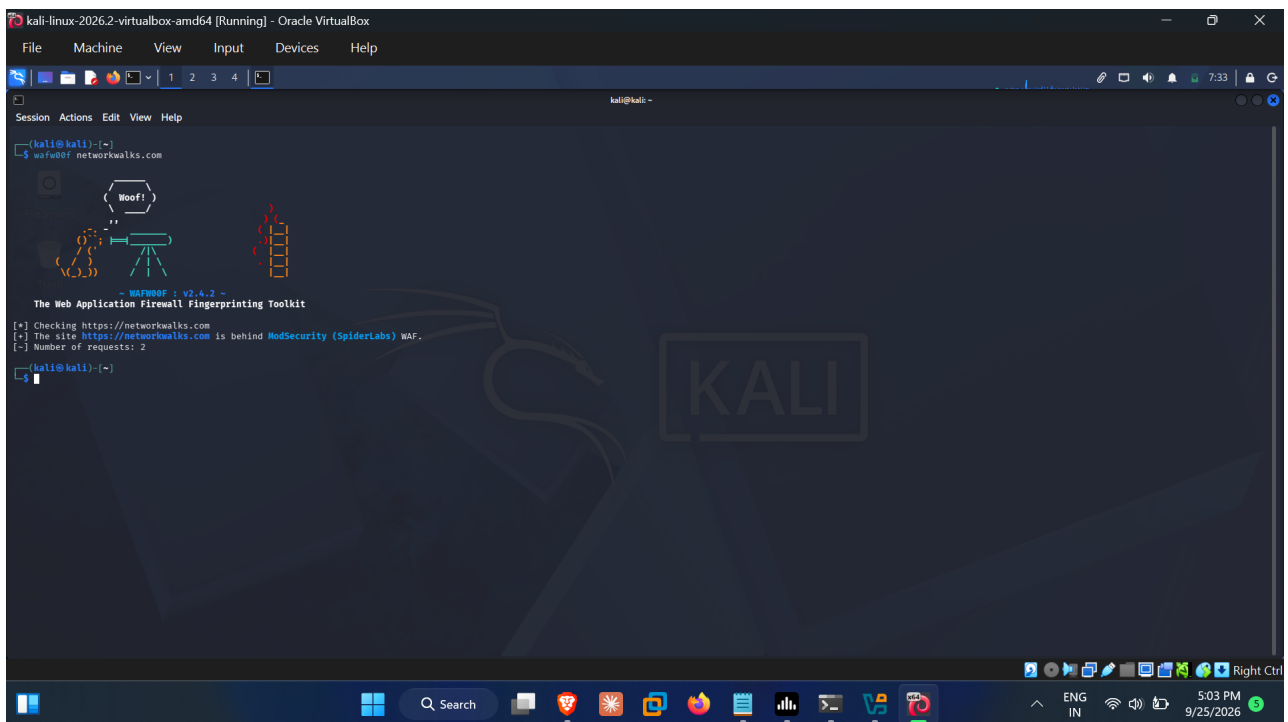
Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

Evidence 3 — nslookup DNS resolution for networkwalks.com.



```
kali@kali:~$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/?rel="https://api.w.org/">, <https://networkwalks.com/wp-json/wp/v2/pages/53?rel="alternate">; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=7b6ae15947411baf40cf677b22cc6d3f; path=/; domain=networkwalks.com; secure; HttpOnly
content-type: text/html; charset=UTF-8
date: Fri, 25 Sep 2026 11:38:00 GMT
server: Apache
```

Evidence 4 — curl HTTP response headers for <https://networkwalks.com>.



```
kali@kali:~$ wafw00f networkwalks.com

  Woof!

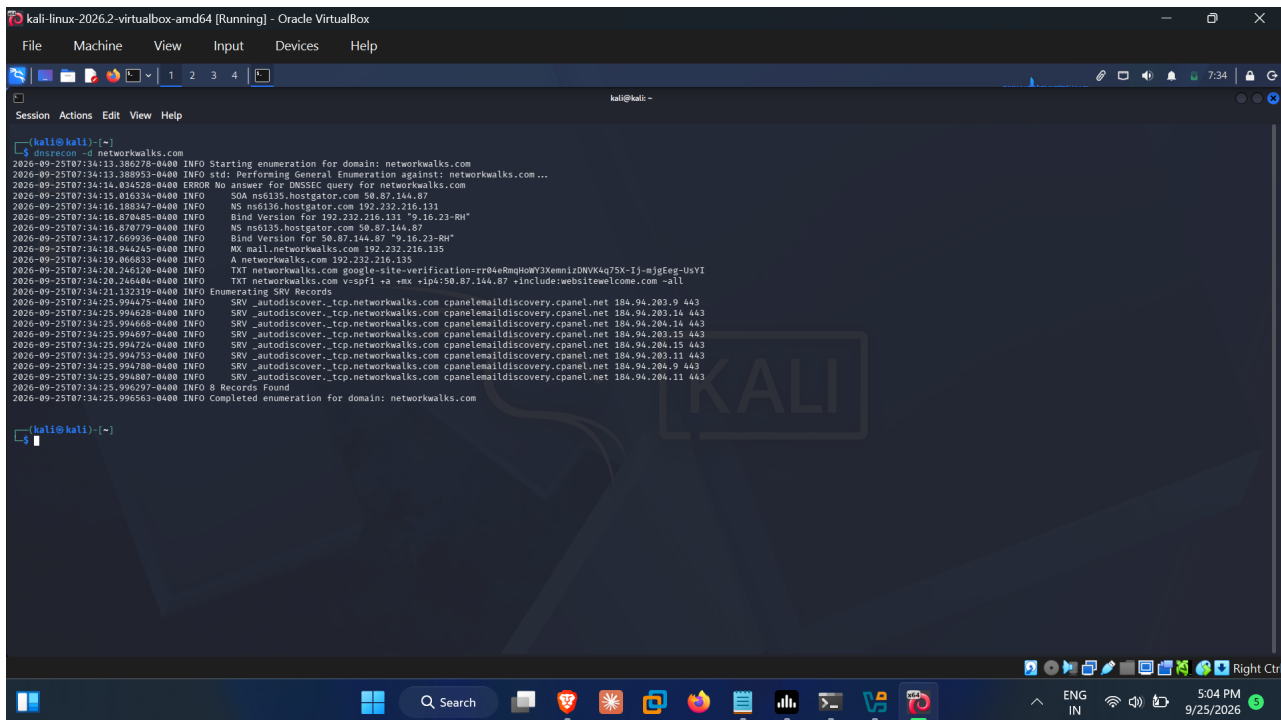
  [Diagram of a web application with a 'Woof!' message and a 'WAF' label]

  ~~~ WAFw00f : v2.4.2 ~~~
  The Web Application Firewall Fingerprinting Toolkit

  [*] Checking https://networkwalks.com
  [*] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
  [*] Number of requests: 2

kali@kali:~$
```

Evidence 5 — wafw00f WAF detection output.

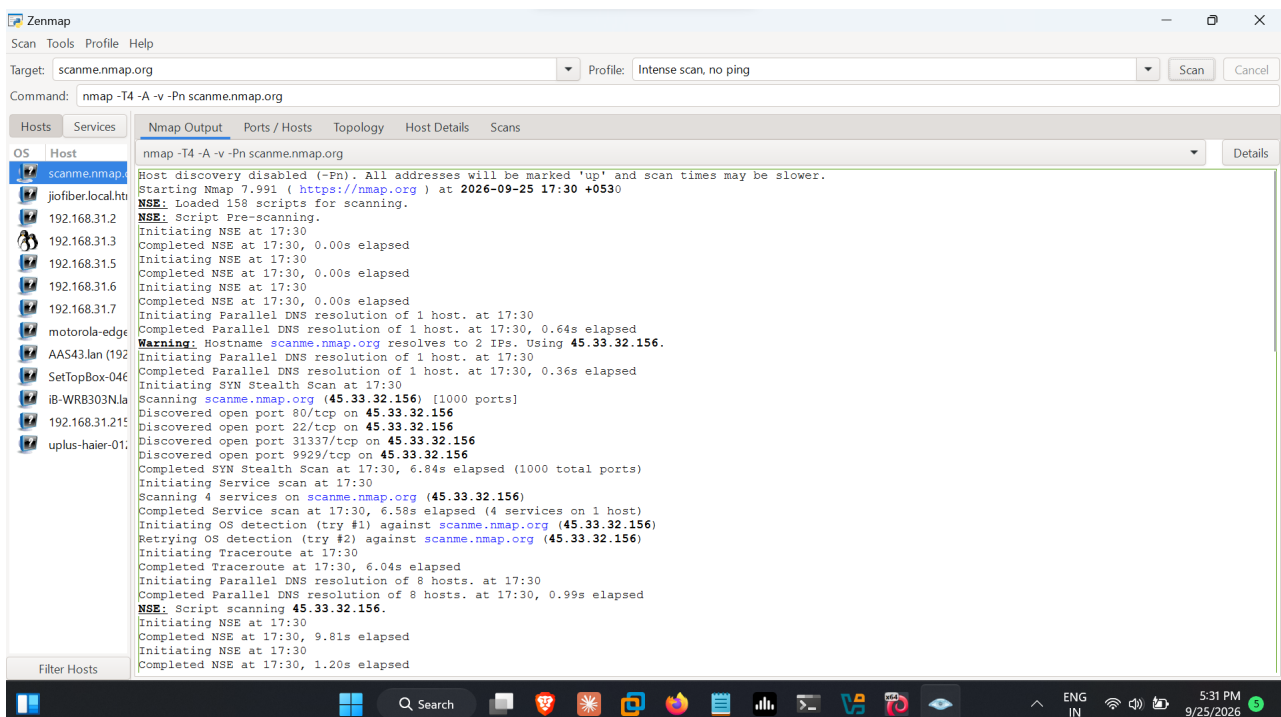


Evidence 6 — DNSRecon enumeration output.

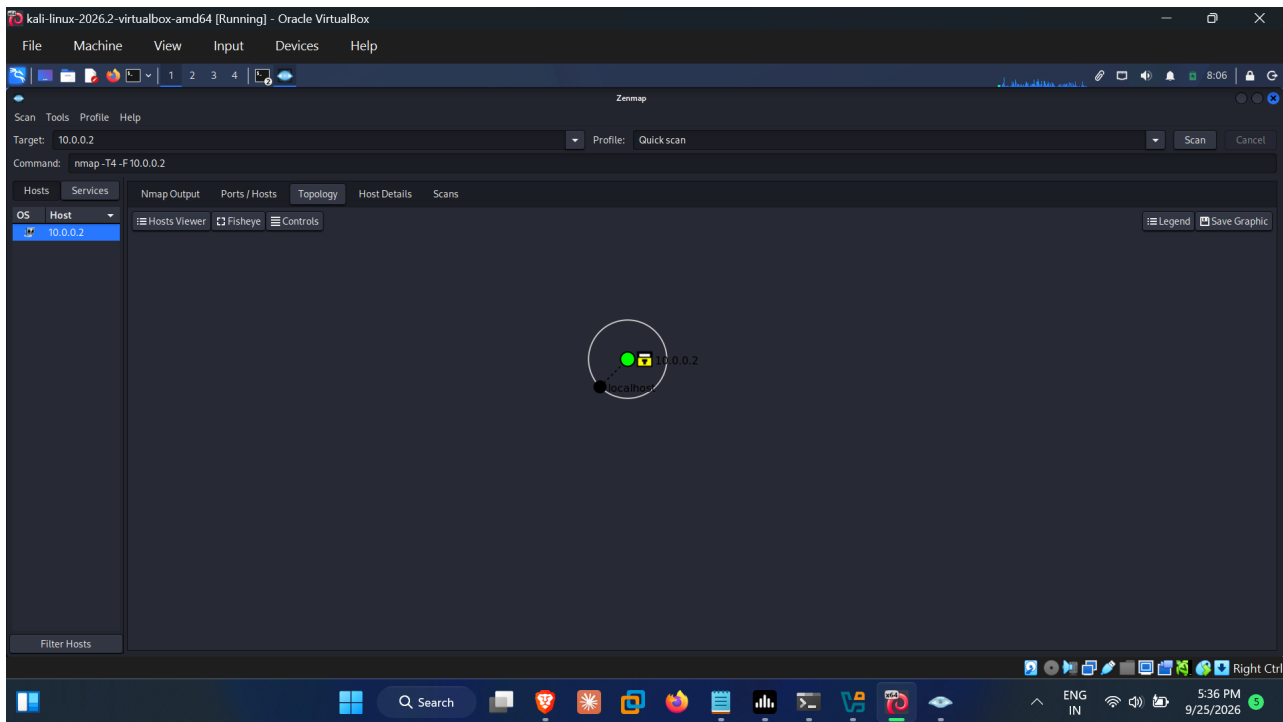
4.2 Network Scanning with Zenmap

I used Zenmap (the graphical interface for Nmap) for controlled network/service discovery. The supplied evidence contains an Intense scan against scanme.nmap.org and a separate Quick Scan/topology view for the local lab target 10.0.0.2.

The scanme.nmap.org evidence resolved the host to 45.33.32.156 and showed four open TCP ports: 80, 443, 31337 and 9929. The second Zenmap evidence shows the local lab topology centered on 10.0.0.2.



Evidence 7 — Zenmap Intense scan of scanme.nmap.org.



Evidence 8 — Zenmap Quick Scan / topology view for the local lab target 10.0.0.2.

5. Risk Analysis / Impact

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress 7.1.2, WordPress Download Manager 3.3.58 and other technologies.	Technology/version information can help an attacker build a target profile and select follow-up checks.	Medium
2	Public server IP identifiable	DNS/WhatWeb resolved networkwalks.com to 192.232.216.135.	Provides infrastructure information about the public web service.	Low
3	HTTP technical information exposed	curl returned HTTP response headers and a WordPress REST API link.	May assist application fingerprinting and further authorized enumeration.	Low
4	WAF technology identifiable	wafw00f identified ModSecurity (SpiderLabs).	Reveals part of the web application's defensive architecture.	Low
5	DNS infrastructure information exposed	DNSRecon returned name-server, mail, TXT/SPF and service-related records.	DNS information can help build a broader infrastructure map.	Medium
6	Network services visible during Zenmap scan	The Zenmap evidence for scanme.nmap.org showed ports 80, 443, 31337 and 9929 open on 45.33.32.156.	Open services increase the observable attack surface and should be reviewed only within authorized scope.	Informational

Risk levels in this report describe the security relevance of the observed information, not confirmed exploitable vulnerabilities. The exercises were primarily information gathering, fingerprinting and network/service discovery. Further authorized testing would be required before treating any observation as a confirmed vulnerability.

6. Recommendations

- Review publicly exposed technology and version information.
- Keep CMS, plugins and supporting web components updated and review security advisories.
- Review HTTP response headers and minimize unnecessary technical disclosure.

- Regularly review DNS records and remove records/services that are no longer required.
- Keep the WAF enabled, properly configured and monitored.
- Perform regular authorized network discovery to identify expected and unexpected devices/services.
- Investigate unknown devices or services found during internal discovery.
- Maintain current network and service documentation.
- Perform reconnaissance and scanning only against systems and networks that are authorized for testing.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning. I used WHOIS to collect domain-registration information, WhatWeb to fingerprint web technologies, nslookup to resolve DNS, curl to inspect HTTP headers, wafw00f to identify WAF technology, and DNSRecon to enumerate DNS records.

For the network-scanning activity, I used Zenmap to observe services exposed by the authorized scan target and to visualize a local lab topology. The work helped me understand how reconnaissance and scanning can build an initial picture of a target environment before any deeper security validation.

I also learned the importance of documenting commands, evidence, observations and security impact clearly. All reconnaissance and scanning activities should remain within an approved scope.

8. Evidences Collected

1. Evidence 1 — WHOIS output for networkwalks.com.
2. Evidence 2 — WhatWeb fingerprinting output for networkwalks.com.
3. Evidence 3 — nslookup DNS resolution for networkwalks.com.
4. Evidence 4 — curl HTTP response headers for https://networkwalks.com.
5. Evidence 5 — wafw00f WAF detection output.
6. Evidence 6 — DNSRecon enumeration output.

9. Submission / LinkedIn

The practical work has also been documented in my LinkedIn learning/progress post. The LinkedIn URL can be added here if required by the Networkwalks submission format.

LinkedIn Post URL: _____

10. Author

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11. Project Information

Program Name: Cybersecurity Program at Networkwalks
Week: 02
Report: Footprinting & Network Scanning
GitHub Repository: _____
LinkedIn Post: _____